PharmaGuard: Hospital Pharmacy Stock and Expiry System

Report 1: Project Description Report

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1. Project Title and Business Idea

Title: PharmaGuard: Hospital Pharmacy Stock and Expiry System.

A hospital pharmacy buys medicines from suppliers, keeps them in stock and gives them to the wards. PharmaGuard keeps all of this information in one place and watches the expiry date of every delivery, so the pharmacy can act before medicine goes to waste.

2. Problem Statement and Purpose

Today, many medicines are thrown away because they expire before they are used. This wastes money and cold-storage space, and sometimes another ward is short of the same medicine at that moment. Nobody sees the problem in time because stock, usage and expiry dates are kept separately.

Purpose: to link each delivery to its supplier, its expiry date and how it is used, to warn staff early when a delivery will probably be left over, and to record when it is moved to a ward that needs it.

3. Project Scope

The system will cover	The system will not cover
Drug and supplier records Receiving deliveries as batches with expiry dates Giving medicine to wards Expiry alerts and how they are closed Moving near-expiry stock to wards Pharmacy staff roles	Patient records and prescriptions Billing and insurance Payments and accounting Other hospital departments Building the real software (this project is design only)

4. Main Classes and Their Responsibilities

Class	Responsibility
Drug	Keeps the details of each medicine: name, category, cost and whether it needs cold storage.
Supplier	Keeps the companies that deliver medicines and how to contact them.
StockBatch	One delivery of a drug. It has its own expiry date and shows how much is left and whether it is at risk.
DispenseRecord	Records every time medicine leaves a batch and goes to a ward.
Ward	A hospital unit that asks for and receives medicines. Its usage shows how much it needs.
Staff (abstract)	Holds what all pharmacy workers share: ID, name, phone, login and dispensing.

Class	Responsibility
Pharmacist	A staff member who reviews and closes expiry alerts.
PharmacyTechnician	A staff member who counts and checks the stock on the shelves.
ExpiryAlert	A warning for a batch that is close to expiry with a lot still unused. It shows the money at risk.
Redistribution	Moves near-expiry stock from the pharmacy to a ward that can use it.

5. System Requirements

Functional requirements

- Register drugs and suppliers.
- Record each delivery as a batch with its received date, expiry date and quantity.
- Record each dispense to a ward, made by a staff member, and reduce the quantity left.
- Create an alert when a batch is within 60 days of expiry and more than 20% is still unused.
- Let a pharmacist review and close alerts.
- Create, approve and complete transfers of near-expiry stock to wards.

Non-functional requirements

- Only logged-in staff can change stock records.
- The system must not allow more to be dispensed than is left in a batch.
- Alerts are checked every day so none are missed.
- It should be simple enough for pharmacy staff with little computer training.